2.1.3 Nucleotides and nucleic acids

Nucleic acids are essential to heredity in living organisms. Understanding the structure of nucleotides and nucleic acids allows an understanding of their roles in the storage and use of genetic information and cell metabolism.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the structure of a nucleotide as the monomer from which nucleic acids are made	To include the differences between RNA and DNA nucleotides, the identification of the purines and pyrimidines and the type of pentose sugar. **PAG10**
(b)	the synthesis and breakdown of polynucleotides by the formation and breakage of phosphodiester bonds	
(c)	the structure of ADP and ATP as phosphorylated nucleotides	Comprising a pentose sugar (ribose), a nitrogenous base (adenine) and inorganic phosphates.
(d)	**(i)** the structure of DNA (deoxyribonucleic acid) **(ii)** practical investigations into the purification of DNA by precipitation	To include how hydrogen bonding between complementary base pairs (A to T, G to C) on two antiparallel DNA polynucleotides leads to the formation of a DNA molecule, and how the twisting of DNA produces its 'double-helix' shape. **PAG9** HSW3, HSW4
(e)	semi-conservative DNA replication	To include the roles of the enzymes helicase and DNA polymerase, the importance of replication in conserving genetic information with accuracy and the occurrence of random, spontaneous mutations. At AS Level learners are **not** required to distinguish between different types of mutation. HSW8
(f)	the nature of the genetic code	To include the triplet, non-overlapping, degenerate and universal nature of the code and how a gene determines the sequence of amino acids in a polypeptide (the primary structure of a protein).
(g)	transcription and translation of genes resulting in the synthesis of polypeptides.	To include, the roles of RNA polymerase, messenger (m)RNA, transfer (t)RNA, ribosomal (r)RNA. HSW8